

Retaining Wall Tutorial

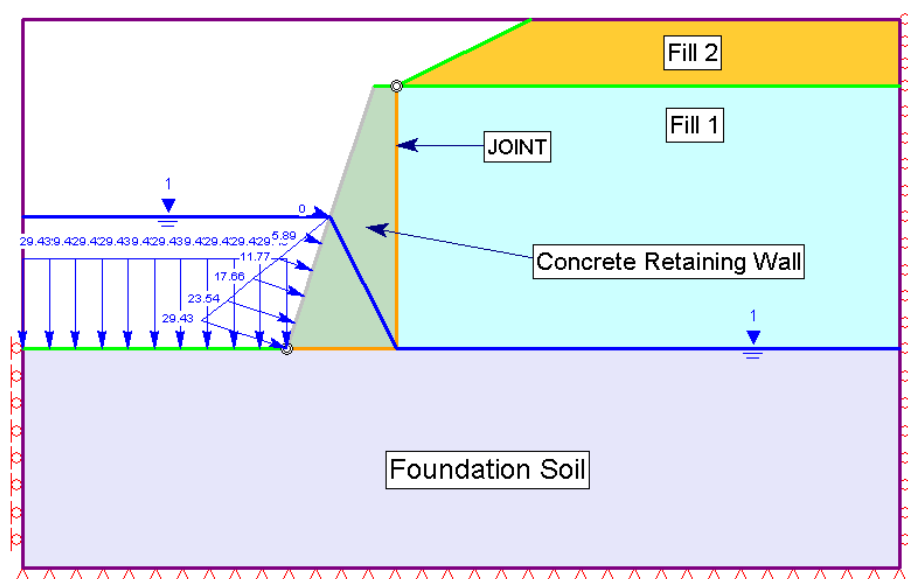
In this tutorial, *Phase2* is used to simulate the construction of an earth retaining wall. The wall is subjected to forces from backfill and from ponded water. Joint elements are included between the wall and the soil so the wall may slip relative to the soil. The model is built in four stages:

1. Bring the foundation soil to equilibrium
2. Add a layer of fill and a retaining wall
3. Add water
4. Add another layer of fill on top of the first

Topics covered

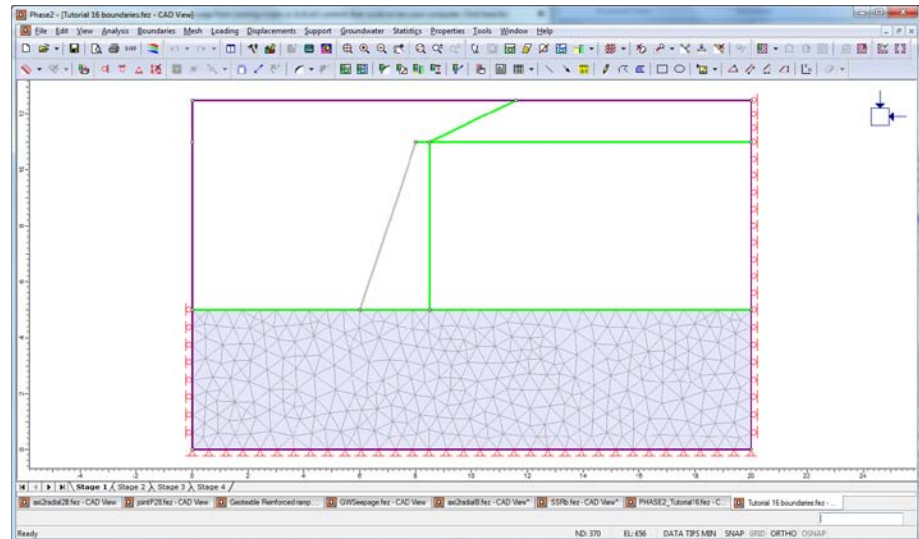
- Joints
- Add vertex
- Staged piezometric lines
- Ponded water load
- Staged loading
- Graph joint data
- Reference stages

Geometry



Model

We will start this tutorial by importing a file in which geometry, materials and boundary conditions have already been assigned. Start the *Phase2* Model program. Select File > Recent Folders > Tutorials Folder and open the file called **Tutorial 16 boundaries.fez**. You will see a model that looks like this.



This is Stage 1 of the model – just the foundation soil. Click through the other stages and you will see the addition of the retaining wall and a layer of fill in stage 2 and another layer of fill in stage 4. In this tutorial we will be adding a joint between the retaining wall and the soil layer and also adding ponded water to the left of the wall.

Add Joint

Click on the tab to show Stage 2. Go to the Boundaries menu and choose Add Joint. If you see a warning that the mesh is going to be reset, click OK to reset the mesh. You will now see the Add Joint dialog. Our joint is man-made and will start and finish at a free surface. Therefore for Joint End Condition, choose the option **Both ends open**. Note that for natural joints found in geological formations you would usually choose both ends closed. We want the joint to be installed at Stage 2 so ensure that the **Install at stage** option is set to 2. The dialog should look like this:

Add Joint

Joint Property: Joint 1

Joint End Condition

☐ Both ends closed
☒ Both ends open
☐ First point closed / last point open
☐ First point open / last point closed

Note: an open joint end should be used if the joint end point intersects a free surface or another joint.

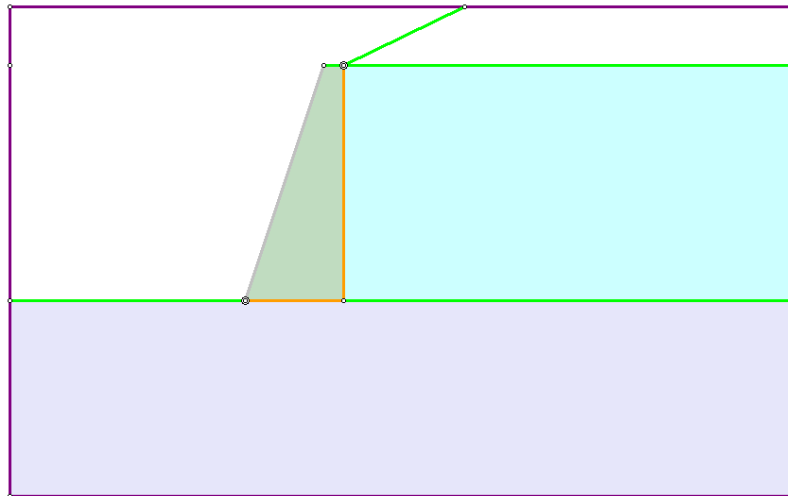
Staging

Install at stage: 2

Note: a Joint removal stage cannot be defined, however a Joint is automatically removed if at least one side of the joint is excavated.

OK Cancel

Click OK. You will now see a cross-hair cursor with which you can select the points that make up the joint. With the mouse, select the point at the bottom left of the retaining wall (6 , 5). The cross-hairs should snap to the existing point. If it does not snap, right click and turn on all of the Snap options. Now select the point at the bottom right of the wall (8.5 , 5) and then at the top of the wall (8.5 , 11). Right click and choose Done. You should now see a joint represented by an orange line as shown.



The open circles at the ends of the joint indicate that it is open at both ends. If you click through the stages then you will see the joint is a light colour in Stage 1 indicating that it is not installed. It is a dark orange (installed) in all other stages.

We now need to set the properties of the joint. Select Define Joints from the Properties menu. For Joint 1, change the criterion to Mohr-Coulomb and the friction angle to 27 degrees as shown.

Define Joint Properties

Joint 1

Name: Joint 1

Color: [Orange]

Slip Criterion

Criterion: Mohr-Coulomb

Tensile Strength (+kPa): 0

Peak Cohesion (+kPa): 0

Peak Friction Angle (deg): 27

☐ Residual Strength

Res. Tensile Strength (+kPa): 0

Res. Cohesion (+kPa): 0

Res. Friction Angle (deg): 35

☒ Initial Joint Deformation

Stiffness

Normal Stiffness (kPa/m): 100000

Shear Stiffness (kPa/m): 10000

Additional Pressure Inside Joint

☐ Groundwater Pore Pressure

☐ Additional Pressure (+kPa): 0

☐ Apply Pressure to Liner Side Only

☐ Stage Joint Properties

Define Factors...

Allow slip starting from stage: 1

Copy To... Statistics... ☐ Show only properties used in model OK Cancel

Click OK to close the dialog. We do not need to assign the Joint 1 properties to our existing joint because it is Joint 1 by default.

Add Piezometric Line

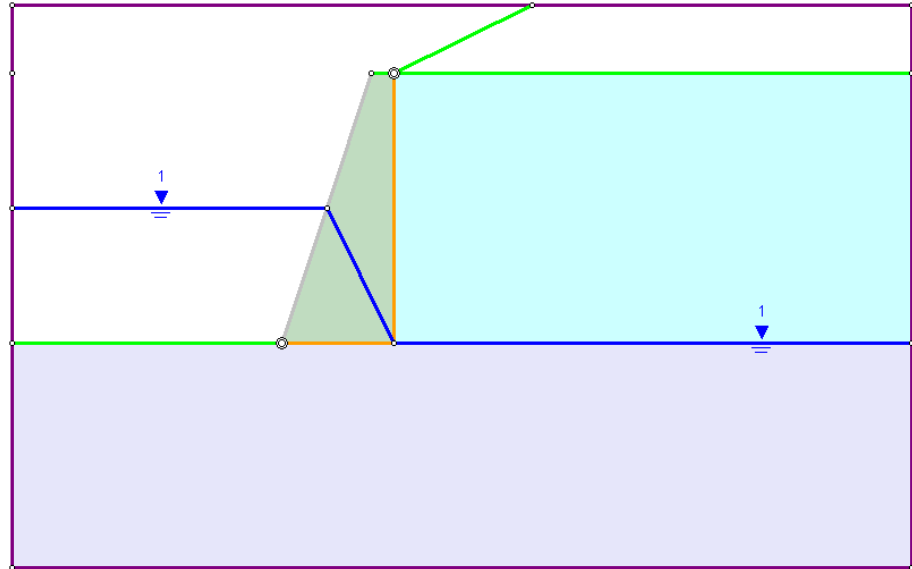
In Stage 3 there will be ponded water to the left of the wall. To draw the piezo line, we should first add a vertex on the wall at the water surface. Go to the Boundaries menu and select Edit > Add Vertices or click the Add Vertices button. The water will be at an elevation of 8 m, so enter the coordinates 7 , 8. Hit Enter. Hit Enter again to finish entering points. You will see a new vertex about half way up the wall on the left side.

NOTE: It is not necessary to add the vertex before drawing the piezometric line, however the new vertex will make adding loads easier later in the tutorial.

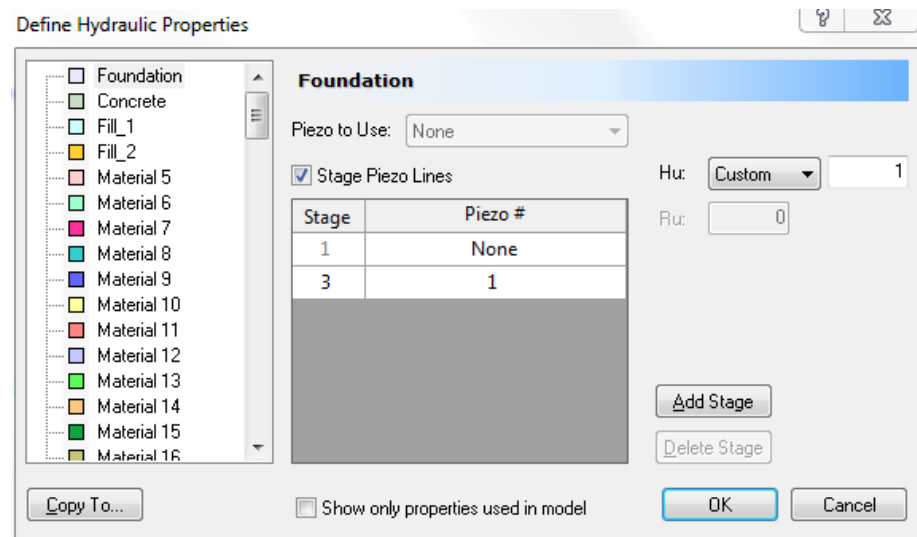
Now select Add Piezometric Line from the Boundaries menu. Enter 0 , 8 for the first point and hit enter. Now click on the new vertex you just created at 7 , 8. Click on the bottom right corner of the wall at 8.5 , 5 and finally click on the top right corner of the foundation soil at 20 , 5. Hit Enter to finish entering points.

NOTE: even though the retaining wall is considered impermeable, the piezo line is defined through the retaining wall so that the pore pressures will be correctly calculated in the foundation soil layer.

You will now see a dialog that lets you choose which materials are affected by the piezometric water level. Select the checkbox next to Foundation and click OK. You should now see the piezometric line as shown.



We only want the water to be added at Stage 3. To do this, select Define Hydraulic from the Properties menu. For the Foundation material, turn on the **Stage Piezo Lines** option. Next to Stage 1 change the Piezo # to none. Stage 2 should now read none as well. Click on Add Stage. Next to Stage 3 change the Piezo # to 1. The dialog should look like this.



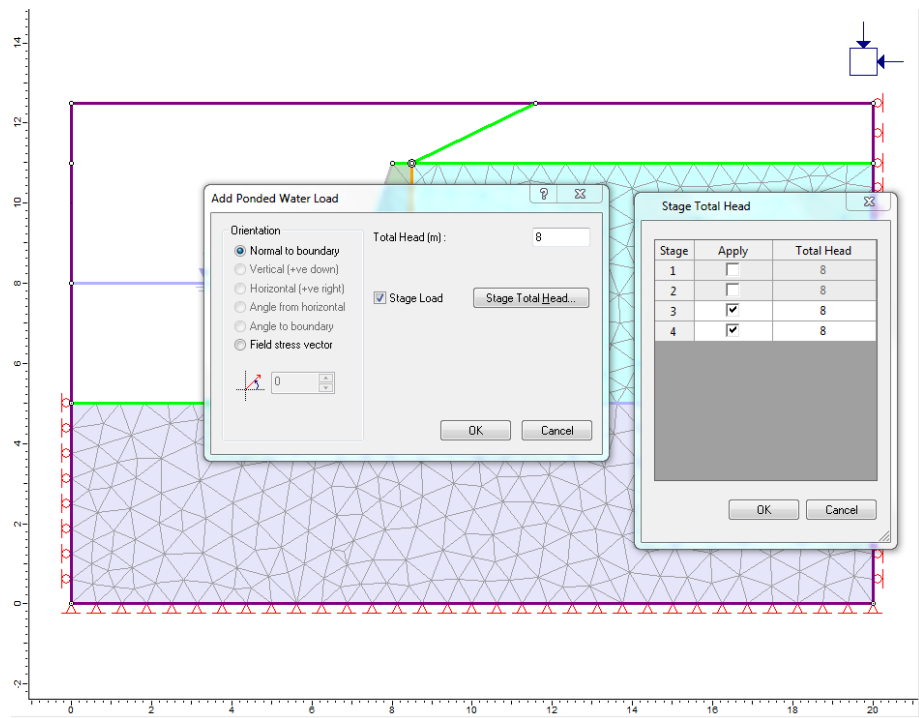
There is no need to set stage 4 since it will automatically be the same as stage 3. Click OK to close the dialog. If you click through the stages you will still see the piezo line plotted at every stage. To change this, you can select **View Piezos by Stage** from the Groundwater menu. Now you should only see the piezo line in stages 3 and 4.

Mesh

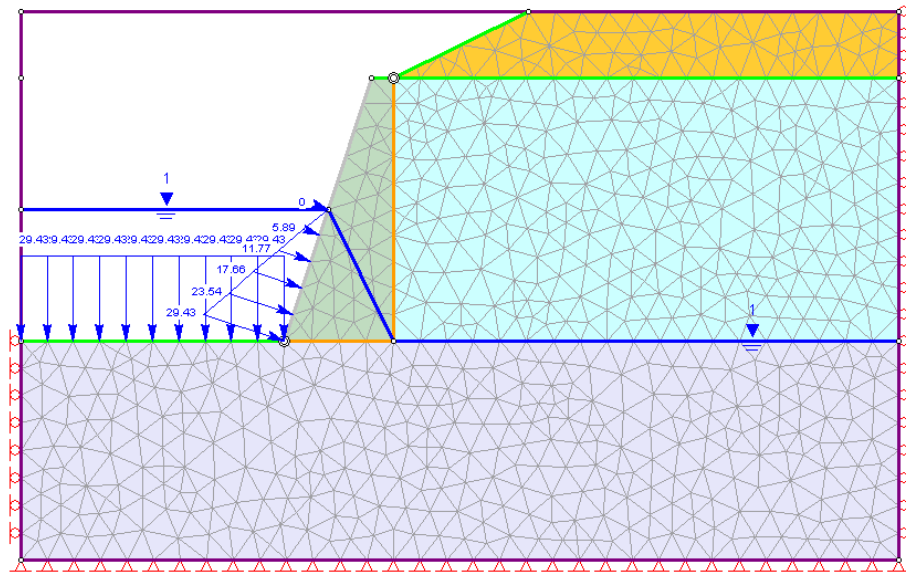
Before we can add the forces caused by the ponded water, we need to generate the mesh. The mesh options are already set up so simply select **Discretize and Mesh** from the Mesh menu.

Distributed Load

The water to the left of the wall will exert a hydrostatic force on the wall and foundation soil. We will simulate this by a distributed load. Go to the **Loading** menu and choose **Distributed Loads**. Select **Add Ponded Water Load** from the sub menu. You will now see a dialog asking for the total head. Enter 8 m. Since we only want the water to be added at stage 3, click on the **Stage Load** option. Click on the Stage Total Head button and unclick the Apply boxes for stages 1 and 2 as shown.



Close both dialogs by clicking OK. You now have to select the boundary segments on which to apply the load. Click the bottom of the pond and the bottom left boundary of the retaining wall below the piezo line. Right click and choose Done. Your model should now look like this for Stage 4.



Note the triangular load applied to the side of the wall. This shows how the hydrostatic force increases with depth.

You have completed the definition of the model. Save the model with a different name using the Save As option in the File menu.

Compute

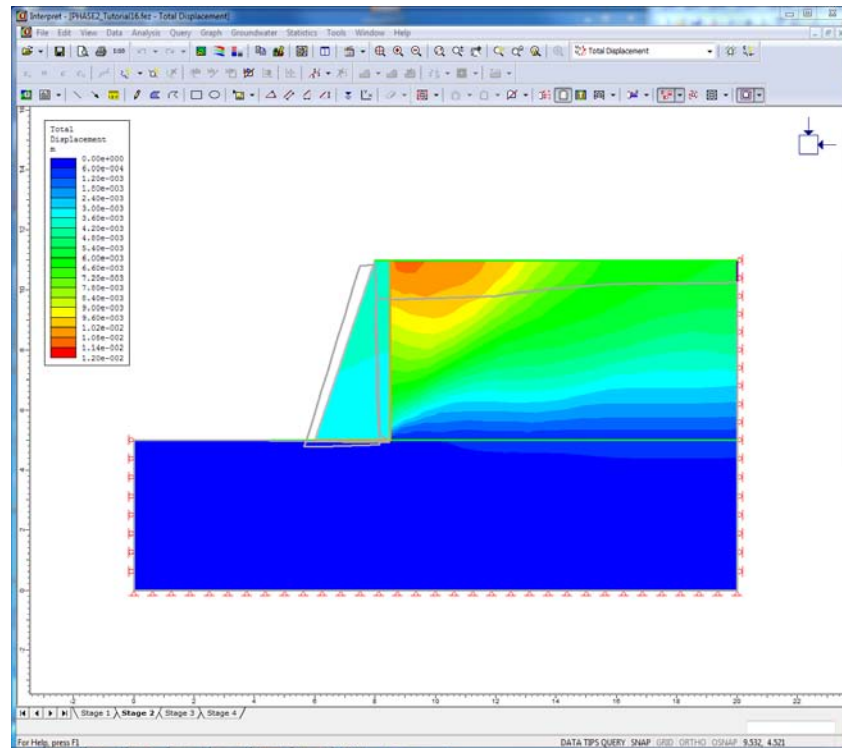
Run the model using the Compute option in the Analysis menu. The analysis should take under a few minutes to run.

Once the model has finished computing (Compute dialog closes), select the Interpret option in the Analysis menu to view the results.

Interpret

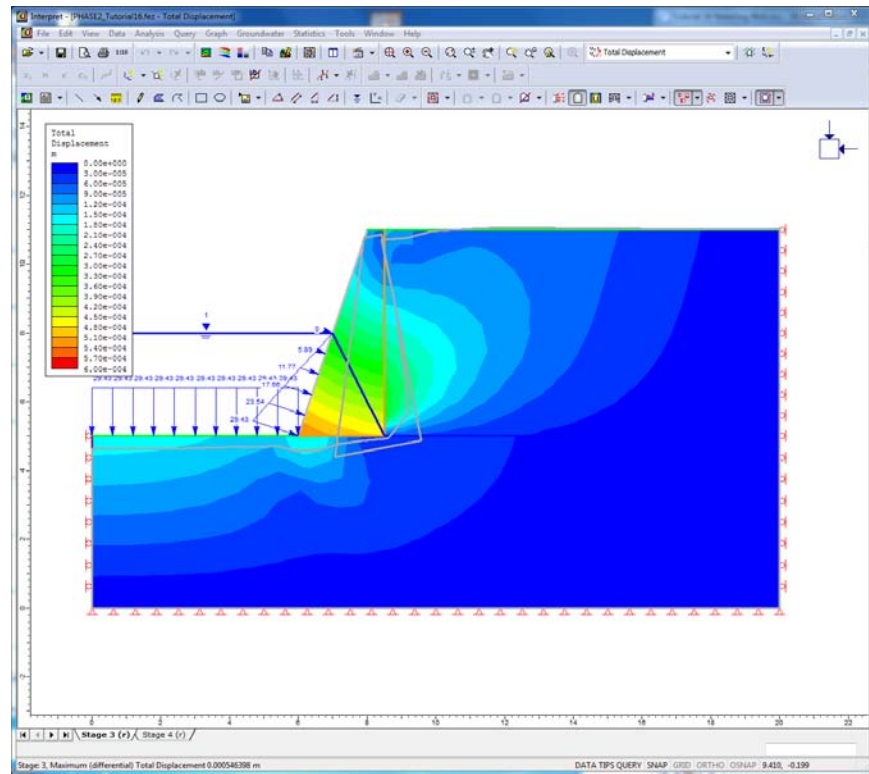
After you select the Interpret option, the Interpret program starts and reads the results of the analysis. You will see the maximum stress in the foundation soil layer for stage 1. Change the display to show contours of Total Displacement. You should see virtually no displacement in the layer since the field stress and body force of the finite elements are in equilibrium in the first stage.

Click on the tab for Stage 2. You will see significant deformation in the fill layer as it settles due to gravity. There is little displacement in the retaining wall since it is made of stiff concrete and does not deform much under gravitational loading. Click on the button to display deformed boundaries. You will see how the retaining wall is being pushed outwards and rotated as shown.



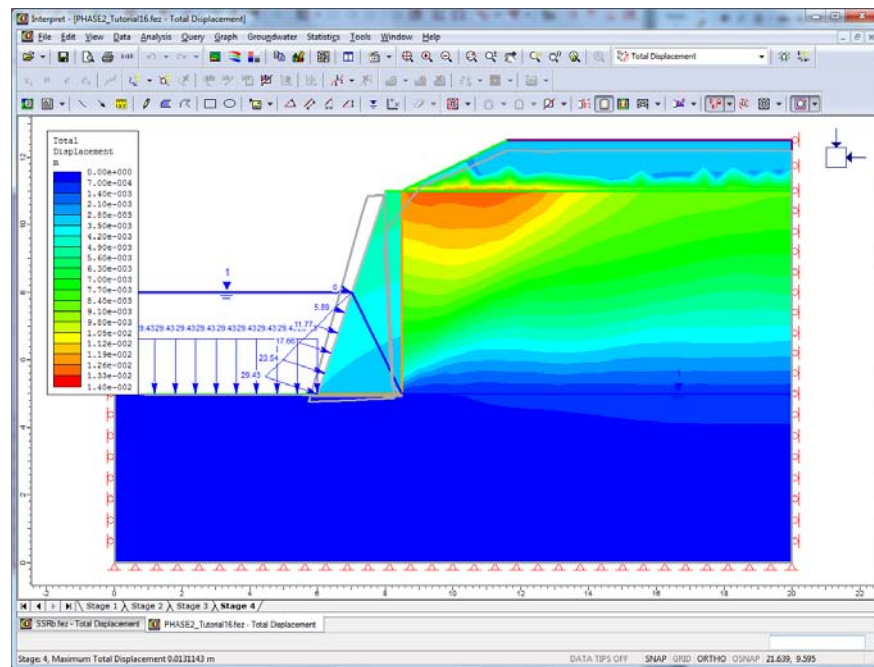
If you click on the button to Display Yielded Joints, you will see all of the vertical joint sections turn red indicating that this entire section of the joint has slipped. It is clear that sliding along the vertical joint is responsible for the displacement contours behind the retaining wall.

Click on the tab to view Stage 3. You can see that the wall is being pushed back to the right slightly due to the force applied by the ponded water. To see this more clearly you can plot the displacement of this stage relative to stage 2. Go to the Data menu and choose Stage Settings. Set the Reference Stage to Stage 2 and click OK. Your plot for Stage 3 will now look like this.



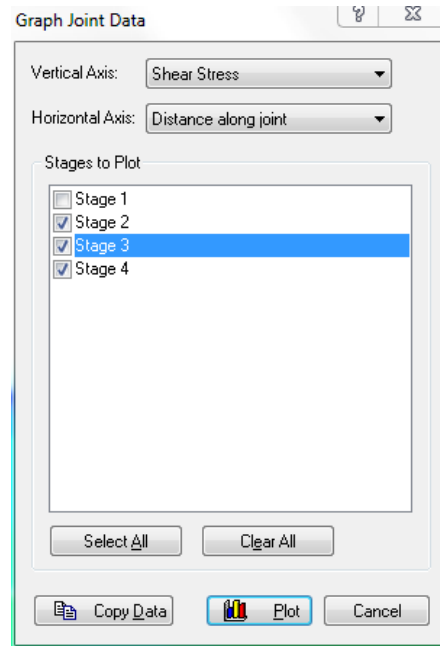
It is clear that the bottom of the wall is being pushed by the water and this is causing displacement and rotation.

Stage 4 shows significant displacement back in the other direction as the second layer of fill is added.

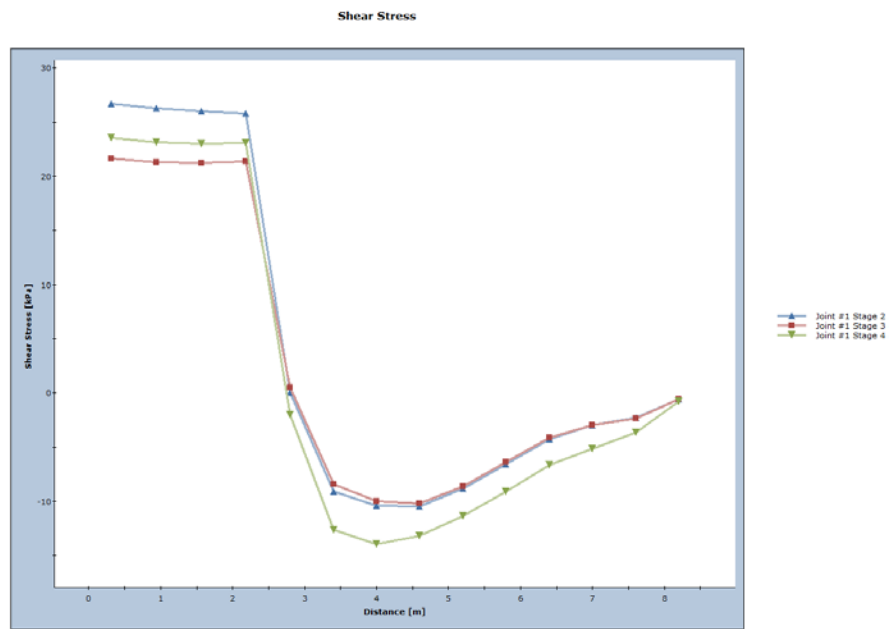


To look at the joint behaviour in more detail, you can graph the joint data. First turn off the reference stage by choosing **Stage Settings** from the **Data** menu and setting the Reference Stage to Not Used.

Select Graph Joint Data from the Graph menu, or simply right click on the joint and choose Graph Joint Data. In the Graph Joint Data dialog, select Shear Stress for the vertical axis and turn on stages 2, 3 and 4 as shown.



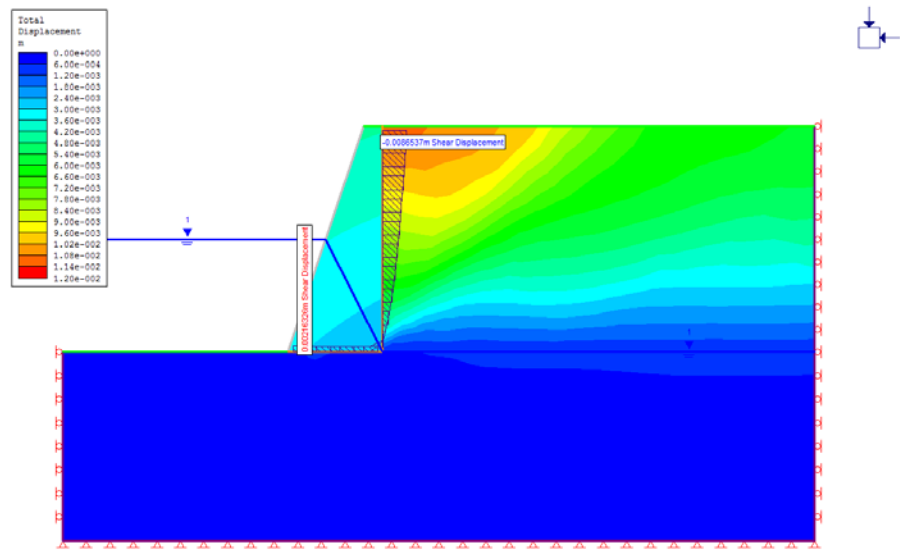
Select Create Plot and you should now see a graph that looks like this.



The first four points show the stress on the horizontal portion of the joint. These show positive values indicating stress causing left lateral motion. You can see how Stage 2 shows the highest shear stress on this segment and how the stress decreases when the water is added in Stage 3. Stage 4 then shows an increase in stress when the extra fill is added. For the vertical section of the joint, the stresses are negative indicating a stress tending to cause right lateral motion.



You may find it easier to plot the joint data directly on the model. To do this, click on the Show Values button on the toolbar. Under the Data heading, turn on the Joints option. For this plot choose Shear Displacement from the pull down menu. Click OK and you will see the shear displacements plotted on the joint. Turn off the deformed boundaries by clicking on the Display Deformed Boundaries button. Turn off the distributed loads by right clicking and choosing Display Options and unchecking the option for Distributed Loads under the Stress tab. You will now see a plot like this for stage 4.



The maximum and minimum values are denoted with red text and blue text respectively. Note the negative (right lateral) slip on the vertical section of the joint as the soil is moving downwards relative to the wall. The bottom of the joint shows positive (left lateral) displacements since the wall is shifting left relative to the foundation.

This concludes the tutorial, you may now exit the *Phase2* Interpret and *Phase2* Model programs.